

Good evening everyone



class start at 8:05

Process A

information about location in disk

Page table

1	-
2	→ f0
3	
4	
5	→ f1
6	
7	→ f3
⋮	

	P1	P2	...
f0	PA	2	
f1	PA	5	
f2	PB	1	
f3	PA	7	
f4	PB	3	
f5			
f6	PB	10	

RAM

Process A	
Pg 1	Process B
2	1
3	2
4	⋮
5	⋮
6	⋮
7	⋮
⋮	⋮

disk

- we have limited memory in ram.
- we create different Page table for each process.
- Page table is stored in ram.

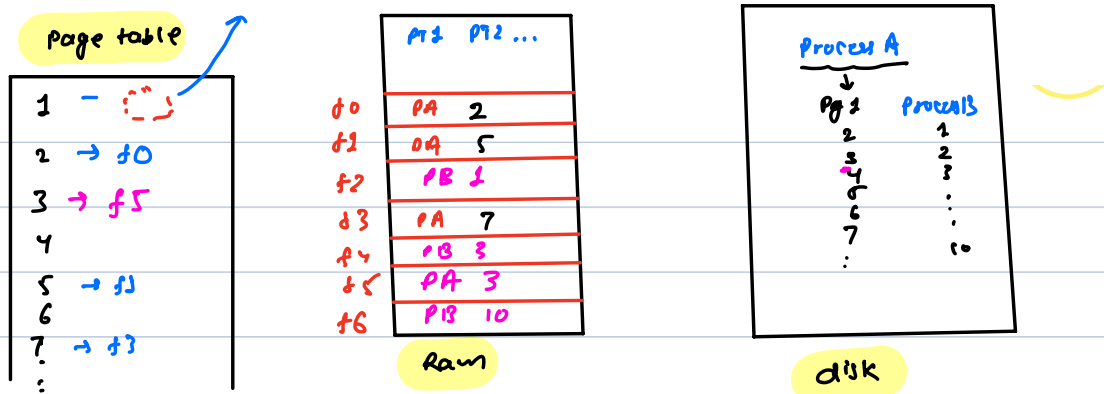
Q1 give me page no. 5 of Process A.

• if data is present
in main memory

• CPU directly retrieve
the data from the
main memory. without
any overhead.
it is referred as a hit.

if data is not
present in ram.

it triggers page
fault.
to manage this
we execute Page
fault handler routine



Q1 give me page no. 3 of process A.

it retrieves data from disk.

then we load the data from disk to ram.

• update the page table of process A.

what if there is no free space?

we use page replacement algorithm

if there is higher page fault -

it leads to degrade performance.

LRU page replacement Algorithm

↓
least recently used

we replace the page that has not been used for longest period of time.

we can only store 3 page of a process

page reference

Process A

1 5 3 4 6 5 3 6 4 6 7 4 5 6

page frame = 3

↑

↑

			3	3	2	5	5	8	4	4	4	4	4	4
Row		5	5	8	6	6	6	6	6	6	6	6	5	5
	1	1	1	4	4	4	3	3	3	3	7	7	7	6
	Pf	Pf	Pf	Pf	Pf	Pf	Pf	hit	Pf	hit	Pf	hit	Pf	Pf

Total no. of page fault = 11

hit = 3

• higher number of page fault lead to thrashing.

• when system spends a significant amount of time swapping pages between disk and ram.

• when you have insufficient amount of space in your ram.

Q if we increase the page frame the page fault is decrease or increase.

Belady's Anomaly

In some page replacement Algorithm specific

FIFO (first in first out) when we increase the number of page frame it may be increase the number of page fault.

FIFO

3 frame

1	4	5	2	1	4	3	1	4	5	2	3	1
PF	PF	PF	PF	PF	PF	PF	hit	hit	PF	PF	hit	PF

[~~1~~ ~~4~~ ~~5~~ ~~2~~ ~~1~~ ~~4~~ ~~3~~ 5 2 1]

total no. of Page fault = 10

hit = 3

4 frame

1	4	5	2	1	4	3	1	4	5	2	3	1
✓	✓	✓	✓	X	X	✓	✓	✓	✓	✓	✓	✓

[~~1~~ ~~4~~ ~~5~~ ~~2~~ ~~3~~ ~~1~~ ~~4~~ 5 2 3 1]

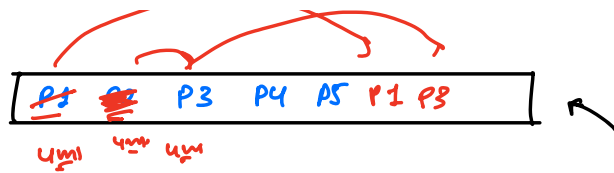
total no. page fault = 11

optimised page replacement algorithm:

remove the page which come last in the future.

round robin

(5)



→ P1 → ~~12ms~~ 8ms

quantum time = 4ms

P2 → ~~3ms~~

P3 → ~~10ms~~ 6ms

P4 → 5ms

P5 → 8ms



AlgoPrep